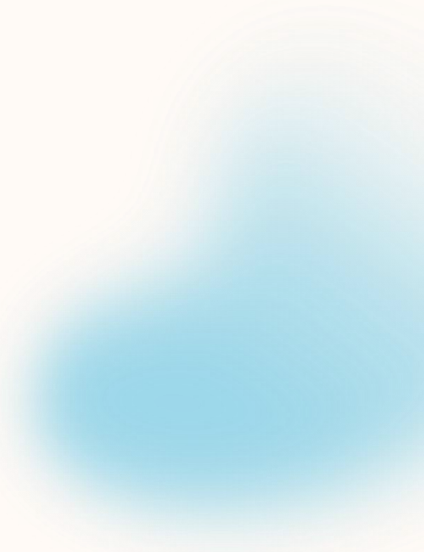


LOOK-OUT TRAFFIC

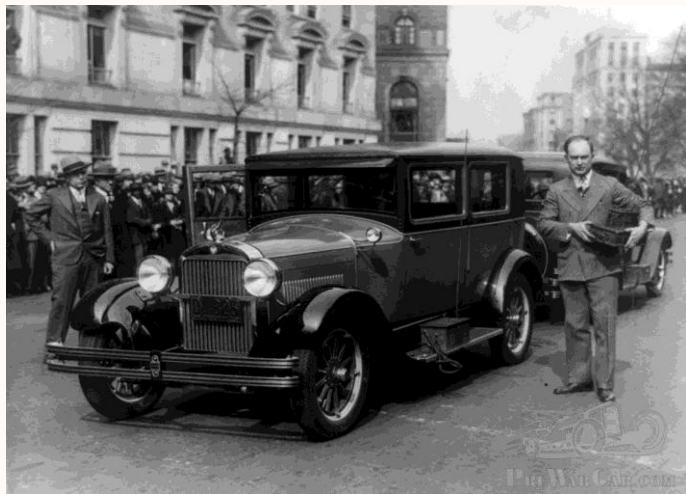
Traffic Sign Image Recognition System using CNN



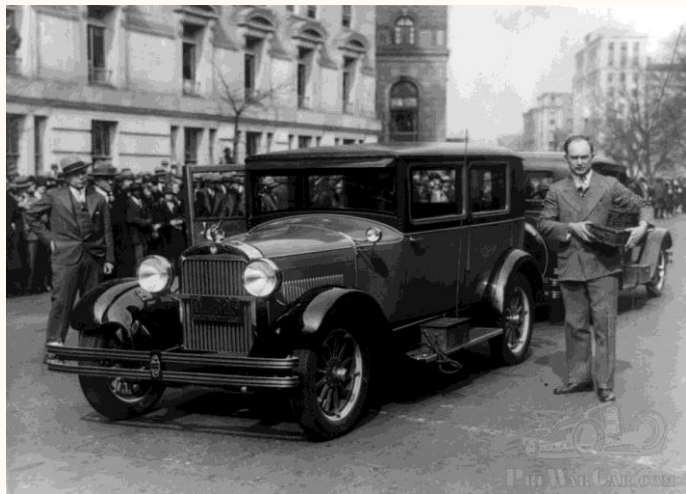
01

Introduction

1925



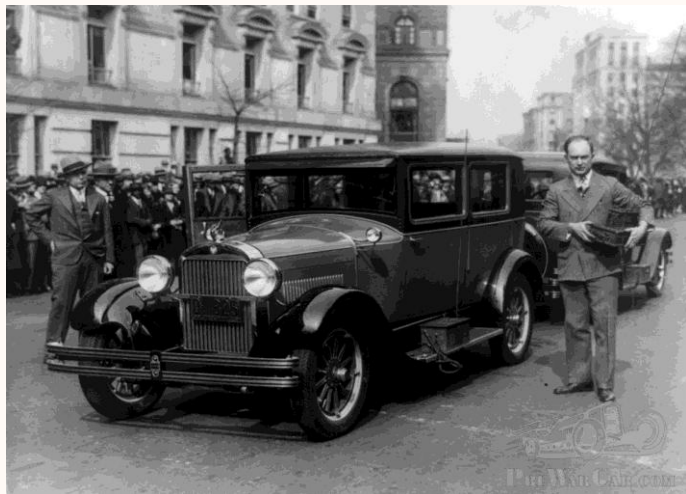
1925



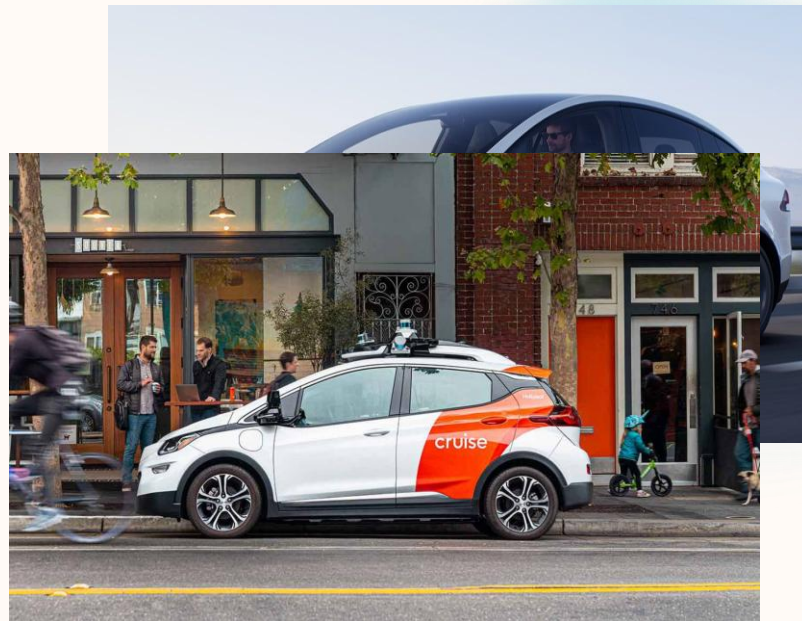
2025



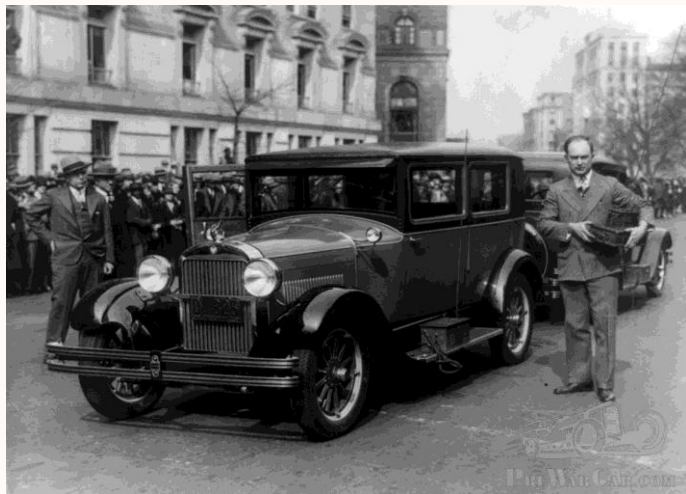
1925



2025



1925



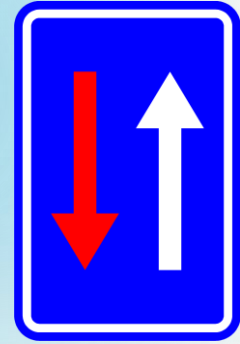
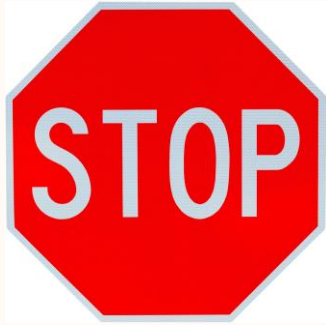
2025





02

Problem Overview



LOOK-OUT TRAFFIC

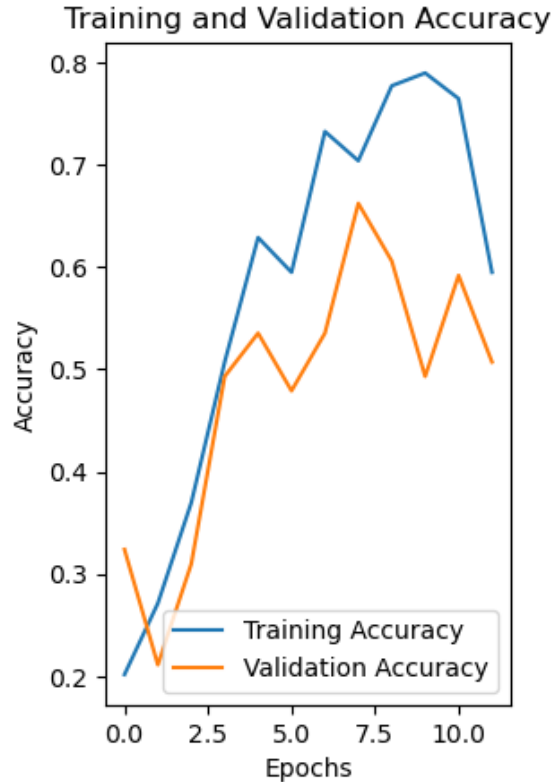




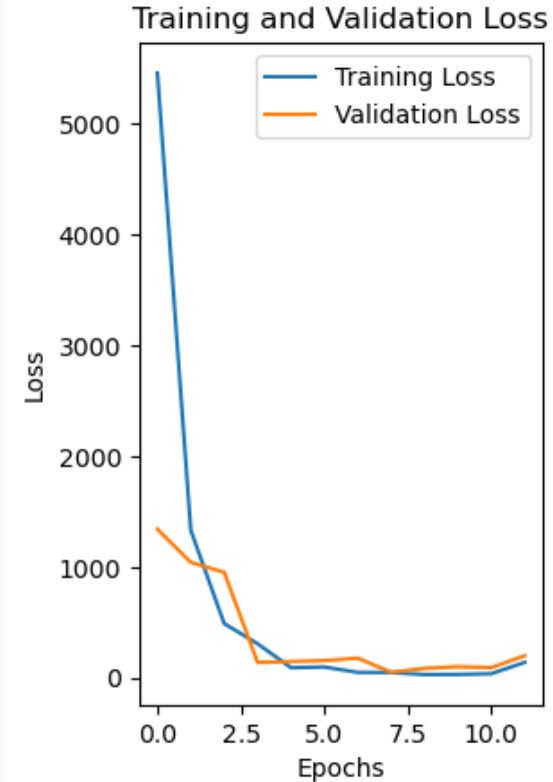
LOOK-OUT TRAFFIC



70-75% Accuracy

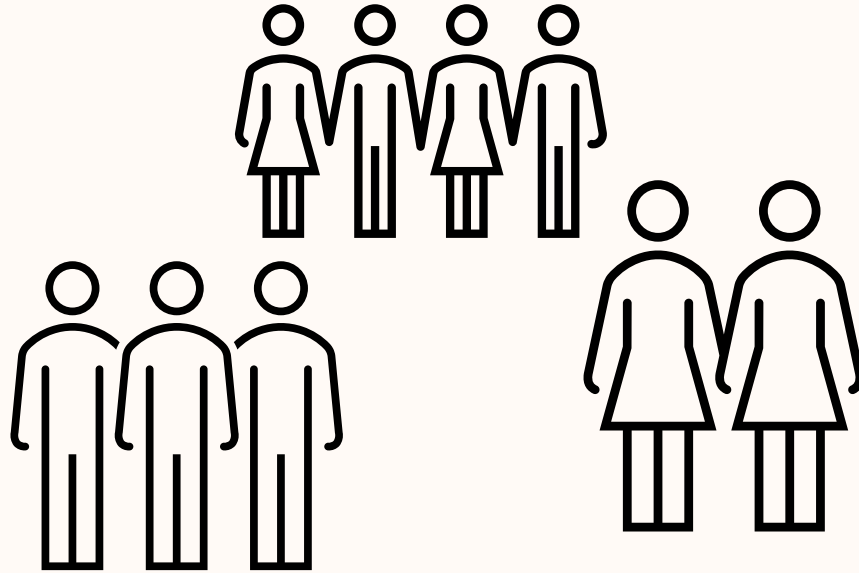


50+ loss



Basic Multilayer Perceptron

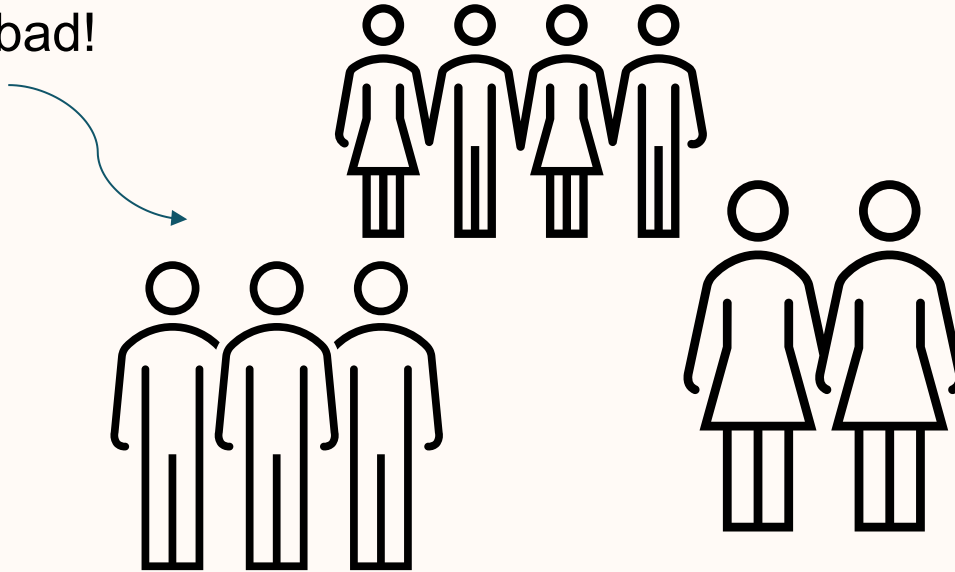
92,5% Accuracy



Human Level Performance

92,5% Accuracy

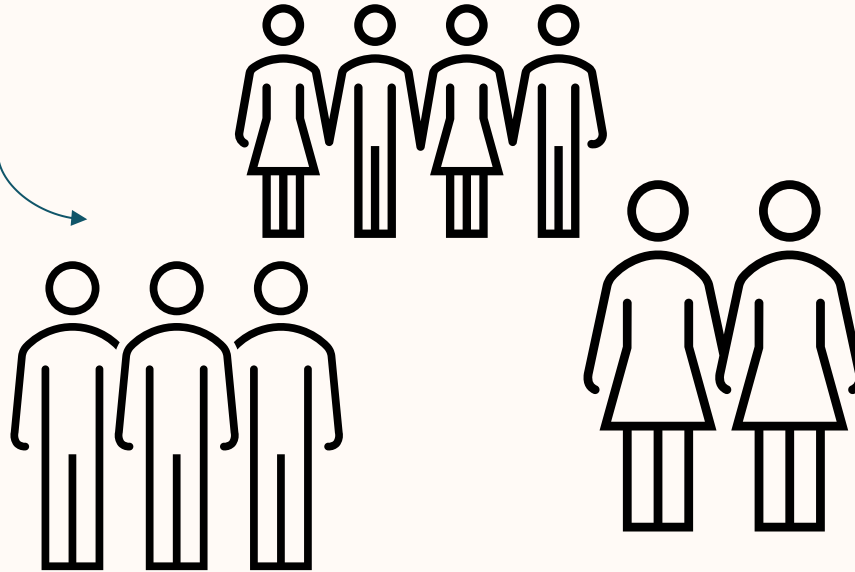
Not too bad!



Human Level Performance

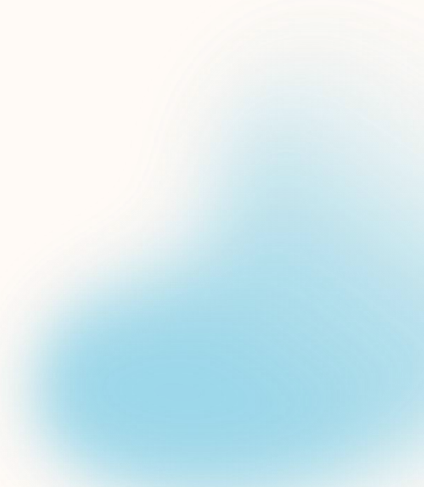
92,5% Accuracy

Not too bad!



But we can do better!

Human Level Performance

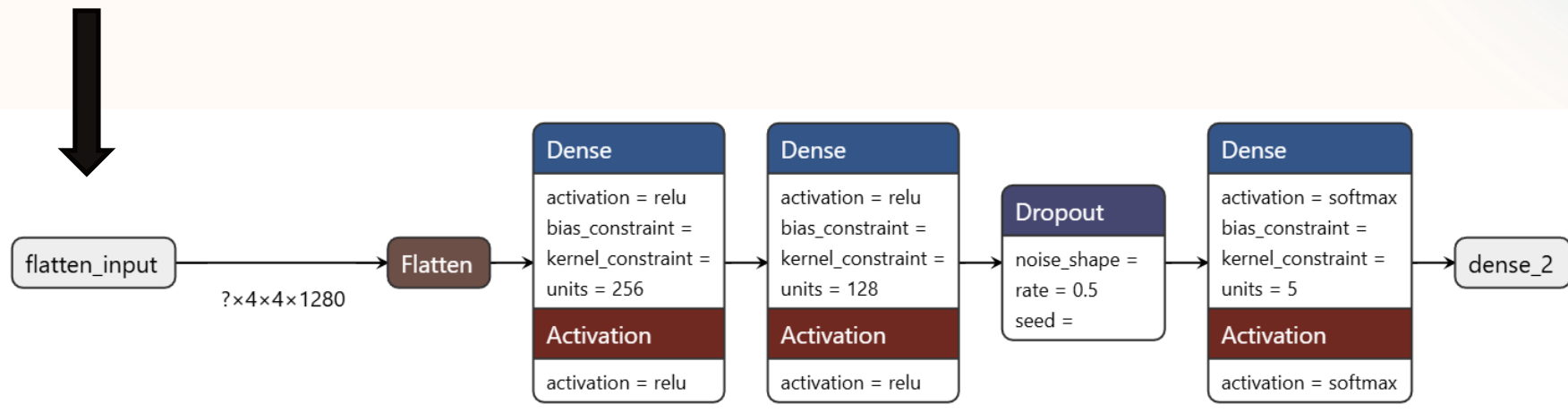


03

Model Overview

Model Architecture

EfficientNetB0 – with pretrained convolutional layers on ImageNet



Training Process

Adam Optimizer

Learning Rate: $1e-4$

- Computation inexpensive
- Quick to learn new tasks
- Handles sparse gradients well

Epochs - 100

Early Stopping

- Callback with a patience of 4
- To avoid overfitting
- To halt training when validation performance stopped improving

Before training, we applied EfficientNet's specific preprocessing, which standardizes the pixel values to match what EfficientNet was originally trained on.

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04

Explainable AI Method

LIME (Local Interpretable Model-agnostic Explanations)

Original Image 1



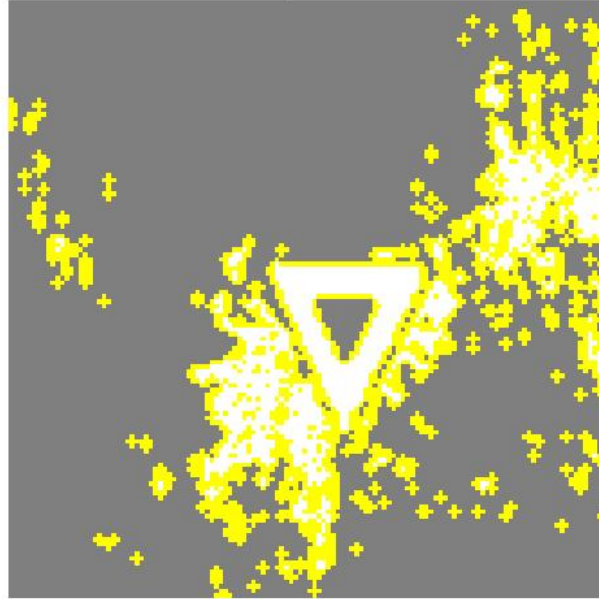
LIME Explanation 1



Original Image 2



LIME Explanation 2



on 1



Original Image 2



Original Image 1



LIME Explanation 2



LIME Explanation 1



Original Image 3



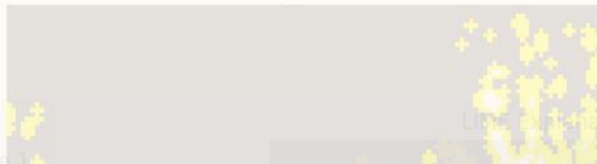
LIME Explanation 3



Original Image 2



LIME Explanation 2



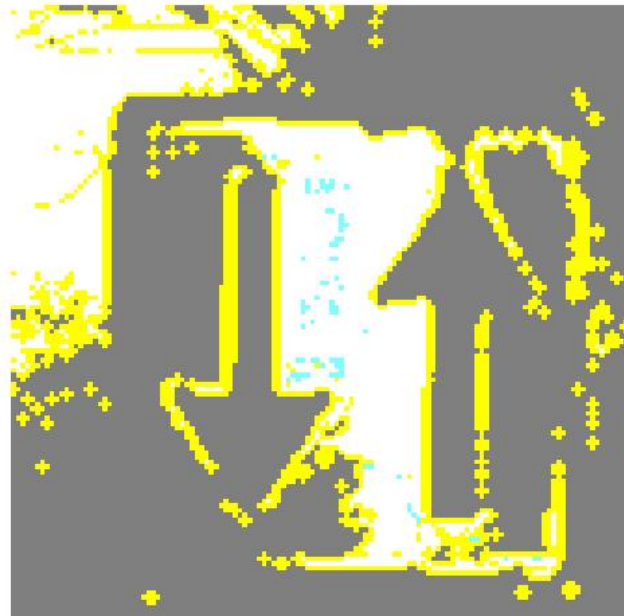
LIME Explanation 1



Original Image 4



LIME Explanation 4

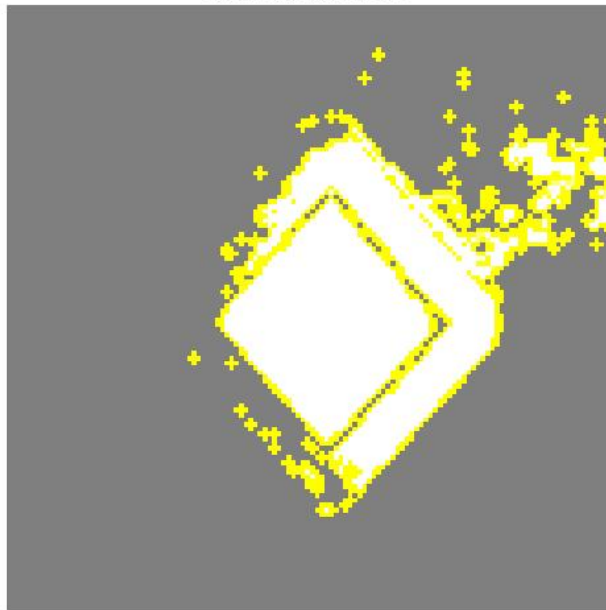


LIME Explanation 2

Original Image 5



LIME Explanation 5



n 4

LIME Explanation 3



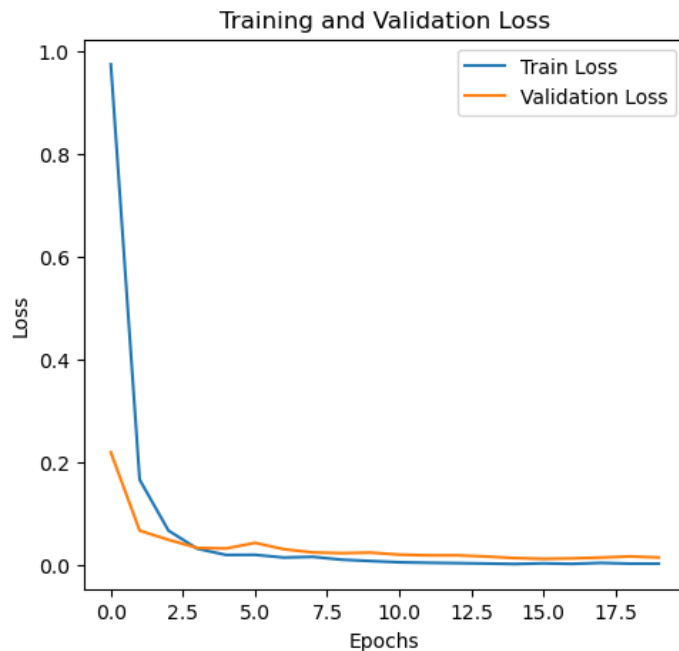
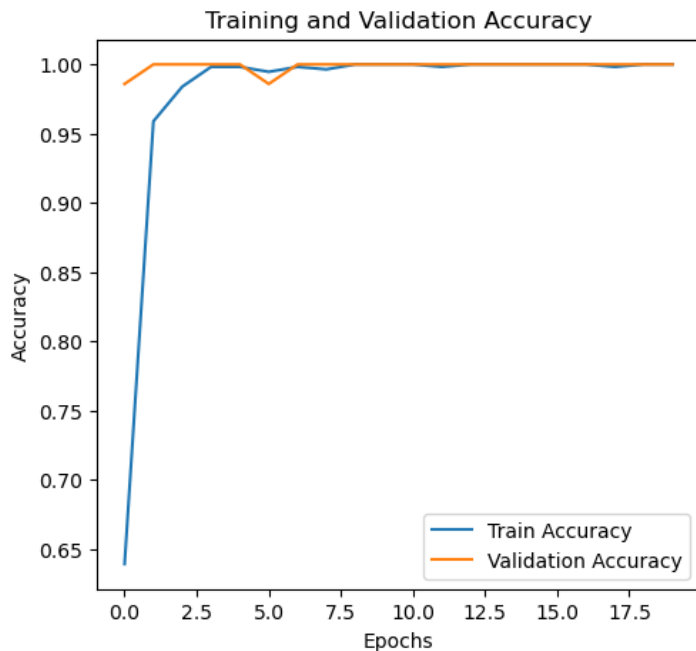


05

Model Results

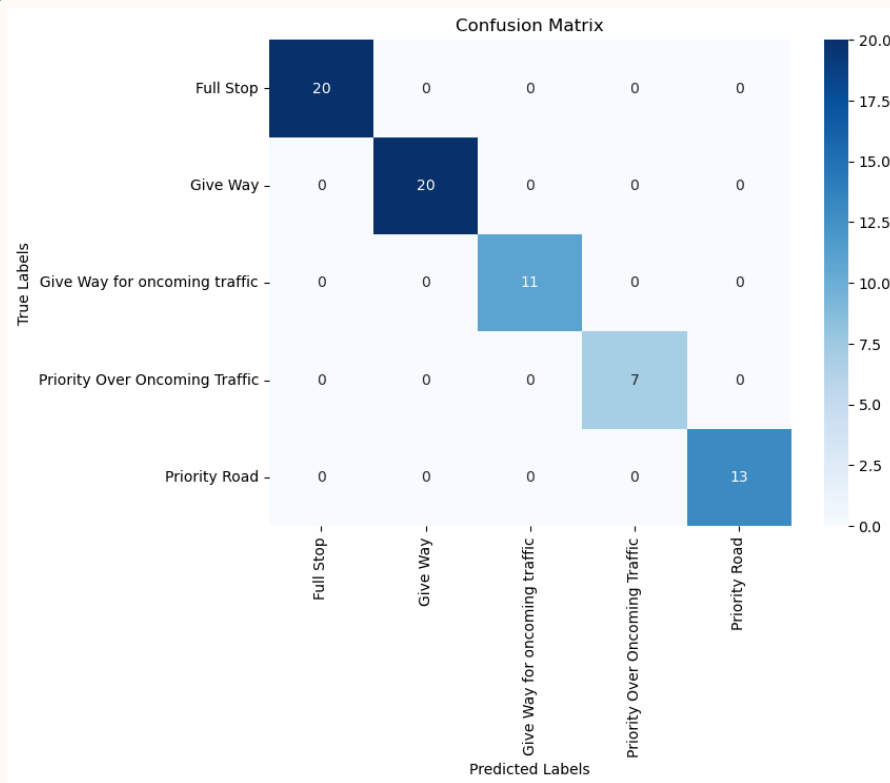


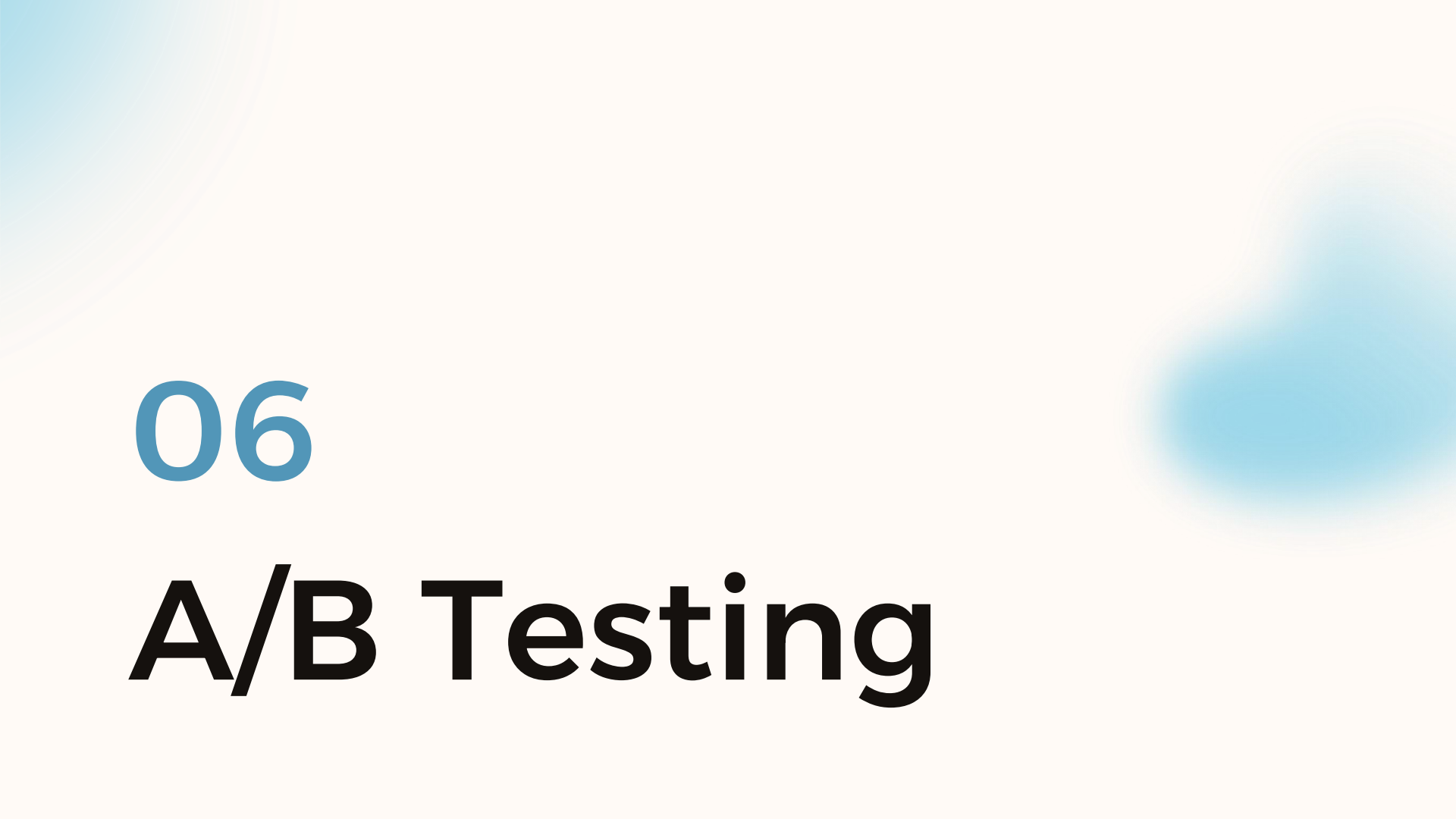
100% ACCURACY





100% ACCURACY





06

A/B Testing

Based on peers' feedback from the DataLab sessions and the Think-Aloud Study,

I hypothesize that adding a visual representation of the model's confidence for each prediction will increase users' trust in the application.

This increase will be measured by higher reported levels of satisfaction and perceived reliability in post-interaction surveys.

LOOK-OUT TRAFFIC



Priority Over Oncoming Traffic

Traffic Rule Explanation: This sign indicates that vehicles on your side of the road have priority over oncoming traffic. Opposing vehicles must yield until your lane is clear.

Recommended Action: Proceed with caution, but maintain your right of way. Be prepared for oncoming vehicles that may not yield. Be aware of other traffic regulations in effect on the road as you drive.



Version A

LOOK-OUT TRAFFIC

Priority Over Oncoming Traffic



Prediction Confidence: 100%

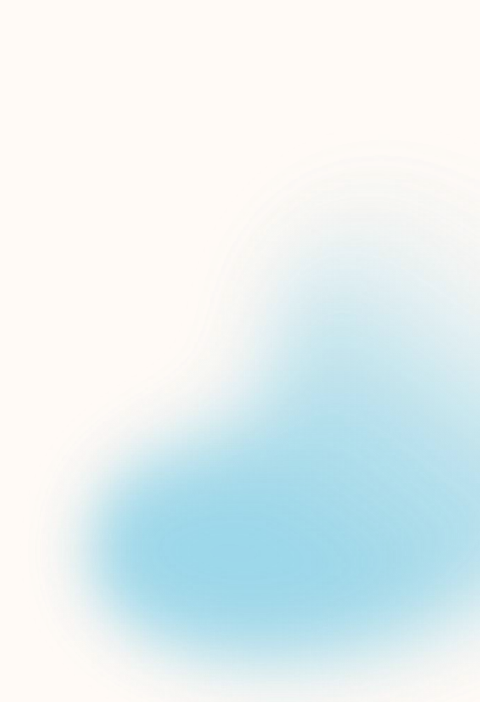
Traffic Rule Explanation: This sign indicates that vehicles on your side of the road have priority over oncoming traffic. Opposing vehicles must yield until your lane is clear.

Recommended Action: Proceed with caution, but maintain your right of way. Be prepared for oncoming vehicles that may not yield. Be aware of other traffic regulations in effect on the road as you drive.



Version B

-	Question	Normality A	Normality B	Homogeneity	T-test p-value	Bootstrapped T-test p-value	Average Test A	Average Test B
0	Question	0.00164	5.66504	0.58753	0.79920	0.79920	5.8077	6.2222
1	Question2	0.00141	5.28505	0.87195	0.32071	0.32071	5.5000	6.1111
2	Question3	0.00499	0.05244	0.17920	0.89760	0.89760	4.7692	5.0556
3	Question4	0.03060	0.00198	0.79613	0.39324	0.39324	5.1538	5.7778
4	Question5	0.00071	0.00020	0.43003	0.01234	0.01234	4.3077	5.9444
5	Question6	0.00196	0.00035	0.30674	0.01642	0.01642	4.4231	5.8889



07

Demo Video

LOOK-OUT TRAFFIC

Seek the innovation

Explore the new image recognition system for
traffic signs

 Take a Picture

or

Choose a traffic sign image...



Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files

Back to main page



08

Conclusions

Future Directions

- Enhancing the quality of the dataset
- Collecting more images
- Keep focusing on gaining users trust through an open interface (not necessarily a confidence level visual)
- Addressing different scenarios

Thank you!

Ana-Maria Farazică | Breda University of Applied Sciences | April 2025